DAVIS–WIELANDT SHELLS OF AFFINE COMPOSITION OPERATORS

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AI assistance was used in adapting and checking the argument and the exposition.

Lahiri, Ojha, Halder, and Birbonshi ask in Question 5.4: "Determine the Davis–Wielandt shell $\mathrm{DW}(C_\varphi)$." Here $\varphi(s) = cs + c_1$, c is a nonnegative integer, and boundedness requires $a = \operatorname{Re} c_1 \geq 0$ for $c \geq 1$ and $a > 1/2$ for $c = 0$.

Lahiri, Ojha, Halder, and Birbonshi, *Numerical range and invariant subspaces of weighted composition operators on the Hardy space of Dirichlet series*, arXiv:2608.29294.

Here $\mathrm{DW}(T) = \{(\langle Tx, x\rangle, \|Tx\|^2) : \|x\| = 1\} \subseteq \mathbb{C} \times \mathbb{R}$.

Answered: exact support-function description. Evidence: Sketch.

For $c = 0$ the shell is the solid ellipsoid below. For $c = 1$ it is the ordinary convex hull $\operatorname{conv}\{(n^{-c_1}, n^{-2a}) : n \geq 1\}$. For $c \geq 2$ and $a = 0$ it is $\{(\lambda, 1) : |\lambda| < 1\} \cup \{(1, 1)\}$. For $c \geq 2$ and $a > 0$ it has the support-function description below, with the origin as the only point of the closure that is not attained.

Proof. When $c = 0$, put $Q = \zeta(2a) > 1$. The operator is rank one: $C_\varphi f = f(c_1)\mathbf{1}$. Its nonzero two-dimensional block is unitarily equivalent to $\left(\begin{smallmatrix} 0 & \sqrt{Q-1} \\ 0 & 1 \end{smallmatrix}\right)$, and it vanishes on an infinite-dimensional orthogonal complement. If a unit vector has coordinates (x, y) on this block, its shell coordinates are $t = Q|y|^2$ and $\lambda = |y|^2 + \sqrt{Q-1}\, y\overline{x}$. Hence

$$Q|\lambda|^2 + t^2 - 2t \operatorname{Re}\lambda - (Q-1)t = Q(Q-1)|y|^2(|x|^2 + |y|^2 - 1) \leq 0.$$

Conversely, for each $0 \leq t \leq Q$, the allowed values of λ form the disk centered at t/Q with radius $\sqrt{(Q-1)(t/Q)(1-t/Q)}$. Choosing $|y|^2 = t/Q$, varying x in the disk $|x| \leq \sqrt{1-|y|^2}$, and placing the remaining norm in the kernel realizes that whole disk. Completing the square shows that the resulting set is exactly

$$\mathrm{DW}(C_\varphi) = \{(\lambda, t) : Q|\lambda|^2 + t^2 - 2t \operatorname{Re}\lambda - (Q-1)t \leq 0\}.$$

For $c = 1$, the monomials $e_n(s) = n^{-s}$ are eigenvectors, with eigenvalues n^{-c_1}. If $f = \sum_n x_n e_n$ is a unit vector, then its shell point is the countable convex combination

$$(\lambda, t) = \sum_{n \geq 1} |x_n|^2 (n^{-c_1}, n^{-2a}).$$

Every such barycenter lies in the ordinary convex hull of the displayed points. Indeed it lies in their closed convex hull; if it is on a relative supporting hyperplane, equality in the averaged supporting functional forces every point of positive weight onto that hyperplane. Repeating in the lower-dimensional affine span reduces to the relative interior, which belongs to the ordinary convex hull. Carathéodory's theorem then represents it using at most four points. Conversely, every finite convex combination is realized by a unit vector with the corresponding squared coefficient moduli.

Suppose next that $c \geq 2$ and $a = 0$. The monomial map fixes e_1 and is the orthogonal direct sum of copies of the unweighted unilateral shift on the remaining chains. It is an isometry, so every shell point has $t = 1$. The numerical range of the unilateral shift is the open unit disk: Cauchy–Schwarz gives $|\langle Sx, x\rangle| \leq 1$, equality would force a unimodular eigenvector, and the unit vectors $x_j = \sqrt{1-r^2}\, r^j e^{ij\theta}$ give every value $re^{i\theta}$ with $0 \leq r < 1$. Combining the fixed vector with the shift copies gives exactly the stated open disk and its fixed point.

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It remains to treat $c \geq 2$ and $a > 0$. Put

$$\mathcal{R}_c = \{n \geq 2 : n \text{ is not a } c\text{-th power}\}.$$

For $0 < r < 1$ and $b \in \mathbb{R}$, define

$$F_{r,b}(z) = \sum_{V,E} (-b)^{|V|}(-1/4)^{|E|} z^{|V|+2|E|} r^{2\sum_{j\in V} c^j + 2\sum_{j\in E} c^j},$$

where V, E run through finite subsets of the nonnegative integers, E has no consecutive elements, and V avoids every endpoint in $E \cup (E+1)$. Define $\tau_r(b)$ to be the first positive real zero of $F_{r,b}$ and set

$$m(b) = \max_{n\in\mathcal{R}_c} \frac{1}{\tau_{n^{-a}}(b)}, \qquad H(\theta, b) = \max\{\cos\theta + b, m(b)\}.$$

Then the shell is

$$\mathrm{DW}(C_\varphi) = \left\{(\lambda, t) : 0 < t \leq 1, \quad \mathrm{Re}(e^{-i\theta}\lambda) + bt \leq H(\theta, b) \text{ for every } \theta, b \in \mathbb{R}\right\}.$$

We justify the scalar series and the support function. On the monomial basis, $C_\varphi e_n = n^{-c_1} e_{n^c}$. Every $n \geq 2$ has a unique representation $n = q^{c^j}$ with $q \in \mathcal{R}_c$ and $j \geq 0$. A diagonal unitary removes the phases of the weights along each chain, giving

$$C_\varphi \simeq 1 \oplus \bigoplus_{q\in\mathcal{R}_c} S_{q^{-a}}, \qquad S_r e_j = r^{c^j} e_{j+1}.$$

For each chain and $b \in \mathbb{R}$, let $J_r(b) = \mathrm{Re}\, S_r + bS_r^* S_r$. Its diagonal entries are br^{2c^j} and its off-diagonal entries are $r^{c^j}/2$. Their absolute sums are finite, so $J_r(b)$ is compact and self-adjoint. For large j, the compression to $\mathrm{span}\{e_j, e_{j+1}\}$ has determinant

$$r^{2c^j}\left(b^2 r^{2c^{j+1}} - 1/4\right) < 0.$$

It therefore has a positive eigenvalue, and the compact operator $J_r(b)$ has a largest eigenvalue $h_r(b) > 0$. Also $h_r(b) \leq r + |b|r^2$.

Let $J_{r,K}$ be the compression to $e_0, \ldots, e_{K-1}$, and let $P_K(z) = \det(zI - J_{r,K})$. Expanding along the last row gives

$$P_0(z) = 1, \quad P_1(z) = z - br^2, \quad P_{j+1}(z) = (z - br^{2c^j})P_j(z) - \frac{r^{2c^{j-1}}}{4} P_{j-1}(z).$$

Extending $J_{r,K}$ by zero,

$$\|J_r(b) - (J_{r,K} \oplus 0)\| \leq r^{c^{K-1}} + |b| r^{2c^K}.$$

Thus the largest eigenvalues of these compressions converge to $h_r(b)$.

The determinant $\det(I - zJ_{r,K})$ expands into exactly the finite choices of V, E above: a nonzero permutation term of a tridiagonal matrix consists of fixed points and disjoint adjacent transpositions. The unrestricted absolute sum is bounded, uniformly on compact z-sets, by

$$\prod_{j\geq 0}(1 + |bz| r^{2c^j}) \prod_{j\geq 0}(1 + |z|^2 r^{2c^j}/4) < \infty.$$

Consequently these finite determinants converge locally uniformly to the entire function $F_{r,b}$. Since the top eigenvalues converge to $h_r(b)$, $F_{r,b}(1/h_r(b)) = 0$. There is no earlier positive zero: for $0 < z < 1/h_r(b)$, every positive eigenvalue η of a finite compression satisfies $z\eta < 1$, while the sum of the absolute eigenvalues is at most

$$B = |b| \sum_{j\geq 0} r^{2c^j} + \sum_{j\geq 0} r^{c^j} < \infty.$$

The negative eigenvalues contribute factors at least one, and $\log(1 - z\eta) \geq -z\eta/(1 - zh_r(b))$ for positive eigenvalues. Therefore

$$\det(I - zJ_{r,K}) \geq \exp\left(-\frac{zB}{1 - zh_r(b)}\right) > 0.$$

Passing to the limit proves $F_{r,b}(z) > 0$ before $1/h_r(b)$, so $\tau_r(b) = 1/h_r(b)$.

For the support direction (θ, b), the self-adjoint operator $\mathrm{Re}(e^{-i\theta}C_\varphi) + bC_\varphi^*C_\varphi$ has top spectral value

$$\max\{\cos\theta + b,\ \sup_{q\in\mathcal{R}_c} h_{q^{-a}}(b)\}.$$

Rotating the basis on each chain removes θ from its shift term. Since $h_{n^{-a}}(b) > 0$ and $h_{n^{-a}}(b) \le n^{-a} + |b|n^{-2a} \to 0$, the supremum is the maximum $m(b)$. This gives $H(\theta, b)$. The vertical support is $0 \le t \le 1$, since C_φ is a contraction with $\|C_\varphi e_1\| = 1$. By Li, Poon, and Sze, *Davis–Wielandt shells of operators*, Theorem 2.3, doi:10.7153/oam-02-20, the Davis–Wielandt shell is convex in dimension at least three; hence these support inequalities determine its closure.

For completeness, the series also gives certified finite approximations. Let $\ell_{n,K}(b)$ be the largest real zero of the above P_K with $r = n^{-a}$, and put

$$L_{M,K}(b) = \max\{0,\ \max_{n\in\mathcal{R}_c,\ n\le M} \ell_{n,K}(b)\}.$$

Then

$$0 \le m(b) - L_{M,K}(b) \le \max\{2^{-ac^{K-1}} + |b|2^{-2ac^K},\, (M+1)^{-a} + |b|(M+1)^{-2a}\}.$$

The shell is compactly generated because C_φ is compact: its images of the monomial basis are orthogonal and have lengths $n^{-a} \to 0$. It is also injective. The inequalities force $\lambda = 0$ when $t = 0$. Indeed, for every unit vector x on a chain and $u > 0$,

$$\langle \mathrm{Re}\, S_r x, x\rangle - u\|S_r x\|^2 \le \|S_r x\| - u\|S_r x\|^2 \le \frac{1}{4u}.$$

Thus $m(-u) \le 1/(4u)$ and $H(\theta, -u) \le 1/(4u)$ for $u \ge 1$. If $(\lambda, 0)$ lies in the closure, its support inequalities give $\mathrm{Re}(e^{-i\theta}\lambda) \le 1/(4u)$ for every $u \ge 1$; letting $u \to \infty$ and varying θ forces $\lambda = 0$.

Every nonzero boundary point of the closure is attained. A supporting functional with nonzero horizontal part has positive support value because $H(\theta, b) \ge m(b) > 0$. A vertical support at the bottom is only the origin; the top support has value $1 > 0$. Let A be the associated compact self-adjoint operator and $h > 0$ its largest eigenvalue. Choose unit vectors whose shell points tend to the boundary point and pass to a weakly convergent subsequence with limit f. Compactness of C_φ and $C_\varphi^*C_\varphi$ gives both the limiting shell coordinates and $\langle Af, f\rangle = h$. Since $\langle Af, f\rangle \le h\|f\|^2$ and $\|f\| \le 1$, we have $\|f\| = 1$, so the point is attained. The relative interior of a closed convex set belongs to the convex set whose closure it is, and the shell is convex; therefore all interior points are attained as well. The origin is not attained because C_φ is injective, but the shell points $(0, n^{-2a})$ for $n > 1$ converge to it. Hence the origin alone is omitted. □